A usage example of code based on OpenMDAO—software framework used to design the Hyperloop

To apply the Hyperloop model, run the file 'src\Hyperloop\ Hyperloop sim.py' in the Hyperloop repository. Each assembly is structured likewise and the following walk-through exhibits the primary layout for setting up a model. The file begins with some library imports and the Input/Output variable definition of the Hyperloop Pod assembly:

```
from openmdao.main.api import Assembly
from openmdao.lib.datatypes.api import Float, Int
from openmdao.lib.drivers.api import BroydenSolver
from openmdao.lib.casehandlers.api import CSVCaseRecorder


from hyperloop.api import (TubeLimitFlow, CompressionSystem, TubeWallTemp,
    Pod, Mission)

class HyperloopPod(Assembly):

    #Design Variables
    Mach_pod_max = Float(1.0, iotype="in", desc="Travel Mach of the pod")
    Mach_c1_in = Float(.6, iotype="in", desc="Mach number at entrance to the first \
        compressor at design conditions")
    Mach_bypass = Float(.95, iotype="in", desc="Mach in the air passing around the pod")
    c1_PR_des = Float(12.47, iotype="in", desc="pressure ratio of first compressor at \
        design conditions")
    PA_tube = Float(99, iotype="in", desc="static pressure in the tube", units="Pa",
        low=0)

    #Parameters
    solar_heating_factor = Float(.7, iotype="in",
        desc="Fractional amount of solar radiation to consider in tube temperature \
        calculations", low=0, high=1)
    tube_length = Float(563270, units = 'm', iotype='in', desc='Length of entire\
        Hyperloop')
    pwr_marg = Float(.3, iotype="in", desc="fractional extra energy requirement")
    hub_to_tip = Float(.6, iotype="in", desc="hub to tip ratio for the compressor")
    coef_drag = Float(.1, iotype="in", desc="capsule drag coefficient")
    n_rows = Int(14, iotype="in", desc="number of rows of seats in the pod")
    length_row = Float(150, iotype="in", units="cm", desc="length of each row of seats")

    #Outputs
    #would go here if they existed for this assembly
    #var_name = Float(default_val, iotype="out", ...)
```

(Source: https://github.com/OpenMDAO-Plugins/Hyperloop)

A pan-India network proposed by Hyperloop One (Source: http://tech.firstpost.com)

This required a huge amount of computer modelling. Ars, a 24-core system owned by the University which limited students to 72 hours of time per project, was thought of as a facilitator. However, since HyperXite was sponsored by Microsoft (among others), the software giant led the undergraduates to a firm called Cycle Computing, which takes computing capacity from cloud services and provides its clients (which are often research teams and private firms) with software that allows them to carry out large-scale simulations in everything from risk modeling for life insurance disbursements to testing potential solar-cell materials. Cycle Computing supported the team in assembling a 128-core cluster on Microsoft Azure to maintain all the necessary data and compute workload.

For actual modeling, HyperXite deployed simulation software ANSYS running the Azure instances, wherein each had 16 Intel Xeon 2.4GHz cores and 7GB RAM per core. This massive computational power enabled the Hyperloop team to model the aerodynamics of the pod as well as the braking system.

Once the design competition was over, the team concluded that the pod shape needed to be changed since the original design weighed too much. It was decided that carbon composites instead of aluminium would be used for pod shell. However, before constructing

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